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AQA GCSE Geography (8035) — Paper 1: Living with the Physical Environment · 3.1.1 The Challenge of Natural Hazards

# 1.1 Tectonic Hazards

## Natural Hazards and Hazard Risk

**Natural hazard** — a natural process or event that poses a threat to people and property (e.g. earthquakes, volcanic eruptions, tropical storms, floods, droughts). An event in an uninhabited area is not a hazard because no-one is at risk.
**Hazard risk** — the probability or chance that a natural hazard will take place and affect people.

| Type of hazard | Cause | Examples |
|---|---|---|
| Tectonic (geological) | Movement of tectonic plates | Earthquakes, volcanic eruptions, tsunamis |
| Atmospheric (weather) | Processes in the atmosphere | Tropical storms, droughts, heatwaves, cold spells |
| Geomorphological | Processes acting on landforms | Floods, landslides |
| Biological | Living things | Wildfires, disease epidemics |

### Factors Affecting Hazard Risk

- **Urbanisation** — dense populations in cities (e.g. Kathmandu, Tokyo) mean more people are exposed. Poor-quality housing in rapidly growing LIC/NEE cities collapses easily.
- **Poverty** — poorer people live in vulnerable locations (steep slopes, floodplains), in badly built homes, and cannot afford protection or insurance. Governments have less money for prediction and response.

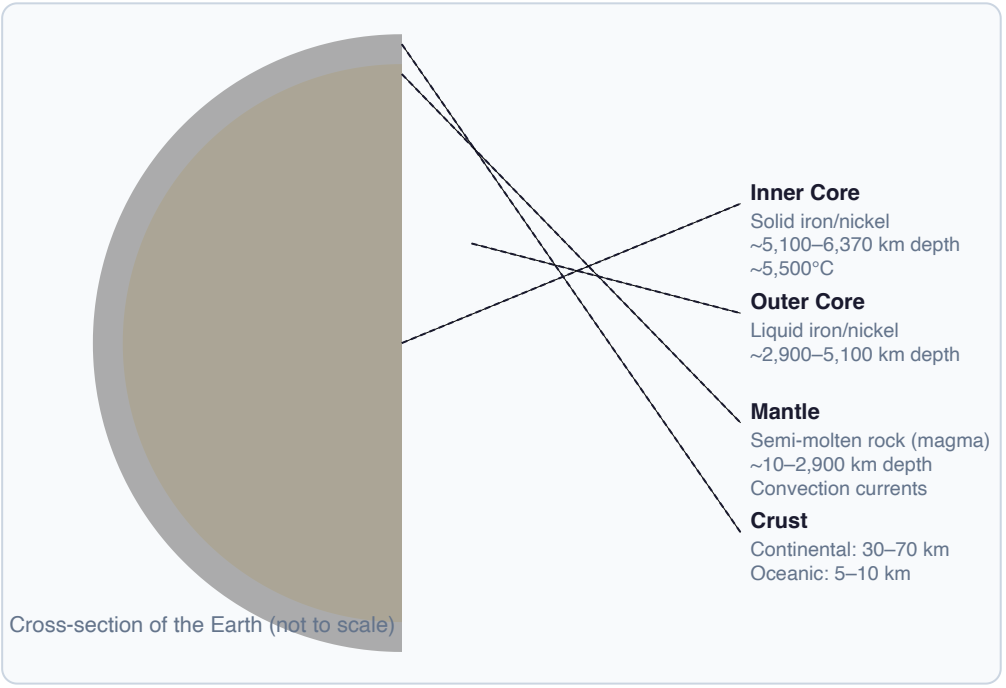
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- **Farming** — fertile volcanic soils and floodplains attract farmers, placing them in hazardous areas.
- **Climate change** — may increase the frequency and intensity of some weather hazards (tropical storms, droughts, floods).

**Exam Tip:** Risk is often summarised as *Risk = Hazard × Vulnerability ÷ Capacity to cope*. The same magnitude of event causes far more deaths where vulnerability is high and capacity to cope is low — this is exactly what the Nepal vs Japan comparison shows.

## 1. Structure of the Earth

The Earth is made up of four concentric layers, each with different properties:



Cross-section of the Earth (not to scale)

| Layer | Thickness | State | Composition | Temperature |
|---|---|---|---|---|
| Crust (Continental) | 30–70 km | Solid | Granite (silica-rich) | ~300–500°C at base |
| Crust (Oceanic) | 5–10 km | Solid | Basalt (denser) | ~200–400°C at base |

| | | | | |
|---|---|---|---|---|
| Mantle | ~2,890 km | Semi-molten (asthenosphere) | Peridotite (silicate rock) | 1,000–3,700°C |
| Outer Core | ~2,200 km | Liquid | Iron and nickel | ~4,000–5,000°C |
| Inner Core | ~1,270 km radius | Solid (high pressure) | Iron and nickel | ~5,100–5,500°C |

**Convection currents** in the mantle are the driving force behind plate movement. Hot magma rises, spreads laterally, cools, then sinks again — dragging tectonic plates with it. This process is called **seafloor spreading** at mid-ocean ridges.

## 2. Plate Tectonics

### Theory of Continental Drift (Wegener, 1912)

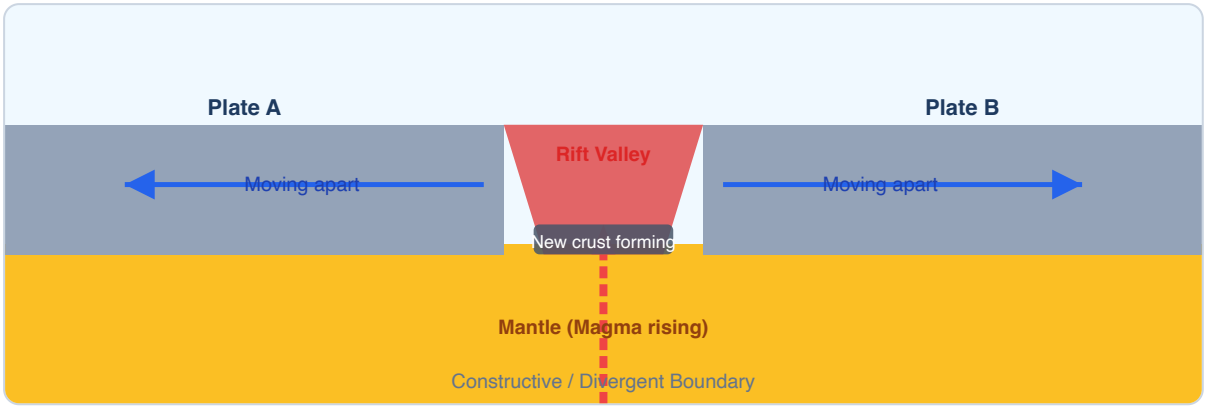
Alfred Wegener proposed that today's continents were once joined as a supercontinent called **Pangaea**. Evidence:

- **Jigsaw fit** — coastlines of S. America and Africa fit together remarkably closely.
- **Fossil evidence** — identical fossils of Glossopteris (a fern) and Mesosaurus (a freshwater reptile) found on opposite sides of the Atlantic — impossible if they had always been separated by ocean.
- **Rock matching** — identical rock sequences found in Brazil and West Africa.
- **Mid-ocean ridges** (later evidence, 1950s–60s) — new oceanic crust forms at mid-ocean ridges as plates pull apart.
- **Paleomagnetism** (later evidence, 1960s) — alternating bands of magnetic polarity on the seafloor either side of mid-ocean ridges prove seafloor is spreading symmetrically.

**Tectonic plates** — rigid segments of Earth's crust and upper mantle (lithosphere) that float on the semi-molten asthenosphere. Major plates: Eurasian, North American, South American, African, Indo-Australian, Pacific (largest), Antarctic. Several smaller plates: Juan de Fuca, Caribbean, Philippine, etc.

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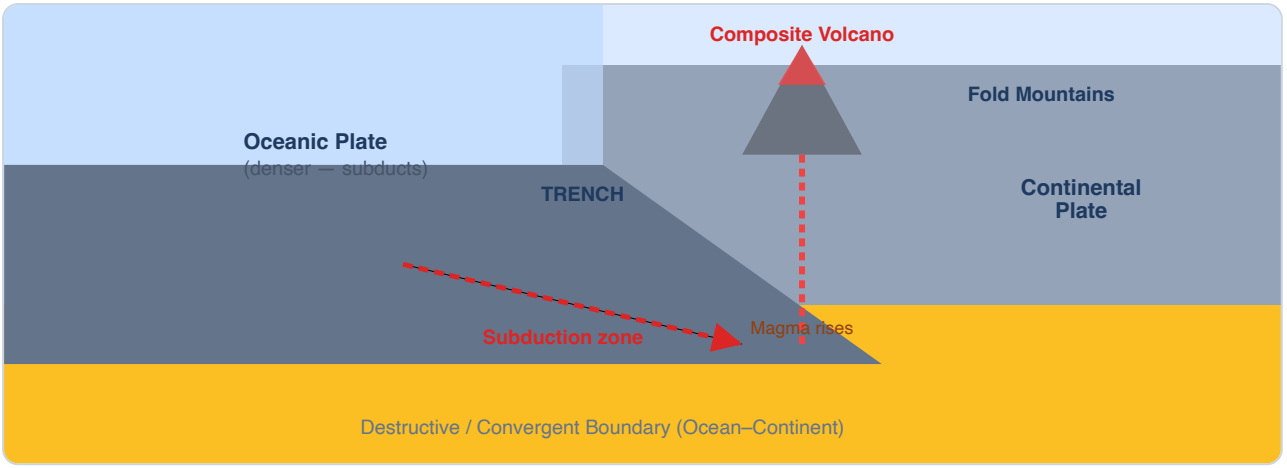
## 3. Constructive (Divergent) Plate Boundaries




- Plates move apart; magma rises through the gap and solidifies, forming **new oceanic crust**.
- Process creates: **mid-ocean ridges** (underwater mountain chains), **rift valleys**, and **shield volcanoes**.
- **Mid-Atlantic Ridge** — Iceland sits on this ridge. Iceland experiences frequent volcanic eruptions (Eyjafjallajokull 2010; Reykjanes Peninsula eruptions from 2021, including near Grindavík in 2023–25) due to its position on the constructive boundary.
- **East African Rift Valley** — continental divergence creating a rift valley (Lakes Tanganyika, Malawi, Turkana).
- **Volcanoes:** Shield volcanoes — wide, gently sloping, basaltic (low-viscosity) lava, less explosive. Example: Iceland's shield volcanoes (e.g. Skjaldbreiður); Mauna Loa, Hawaii, is a hot-spot shield volcano.
- **Earthquakes:** Generally shallow focus and low magnitude — less destructive.

## 4. Destructive (Convergent) Plate Boundaries

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## Ocean-Continent (Subduction)

- Denser oceanic crust subducts beneath lighter continental crust into the mantle — forms a **subduction zone** and **deep-sea trench**.
- Oceanic crust melts in the Benioff zone; magma is forced upward through the continental crust, forming **composite (stratovolcanoes)**: steep, explosive, andesitic lava (high viscosity). Examples: Andes (Chile, Peru), Cascade Range (Mt St Helens, USA — erupted 1980), Mt Pinatubo (Philippines — erupted 1991).
- Earthquakes: deep focus (Benioff zone), high magnitude — very destructive.

## Ocean-Ocean Convergence

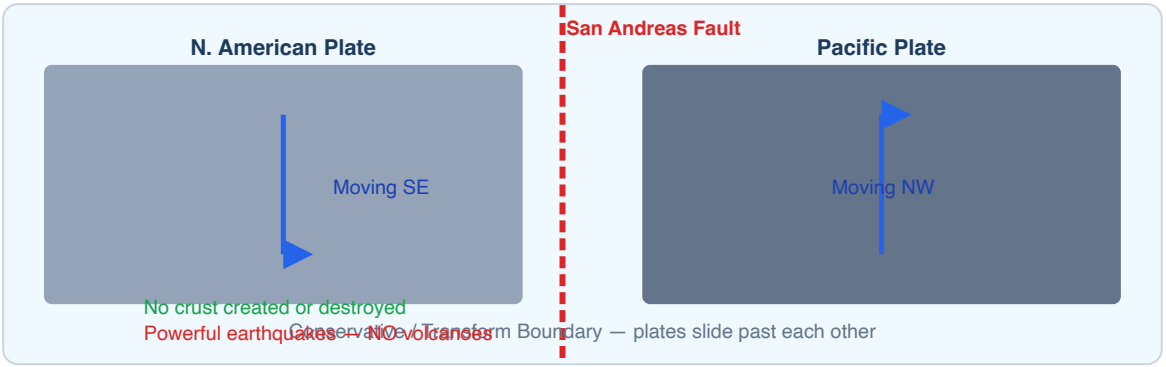
- One oceanic plate subducts under the other. Forms: deep-sea trenches (Mariana Trench — deepest point on Earth, 11,034m) and **island arcs** — chains of volcanic islands (Japan, Caribbean islands).

## Continent-Continent Convergence

- Neither plate subducts (both are too buoyant). Instead, crust crumples and folds upward to form **fold mountain ranges**. No volcanoes. Powerful earthquakes.
- Example: **Himalayas** — Indo-Australian plate colliding with Eurasian plate. Still rising ~5mm/year. Nepal Earthquake 2015 a result of this collision zone.
- Alps — African plate colliding with Eurasian.

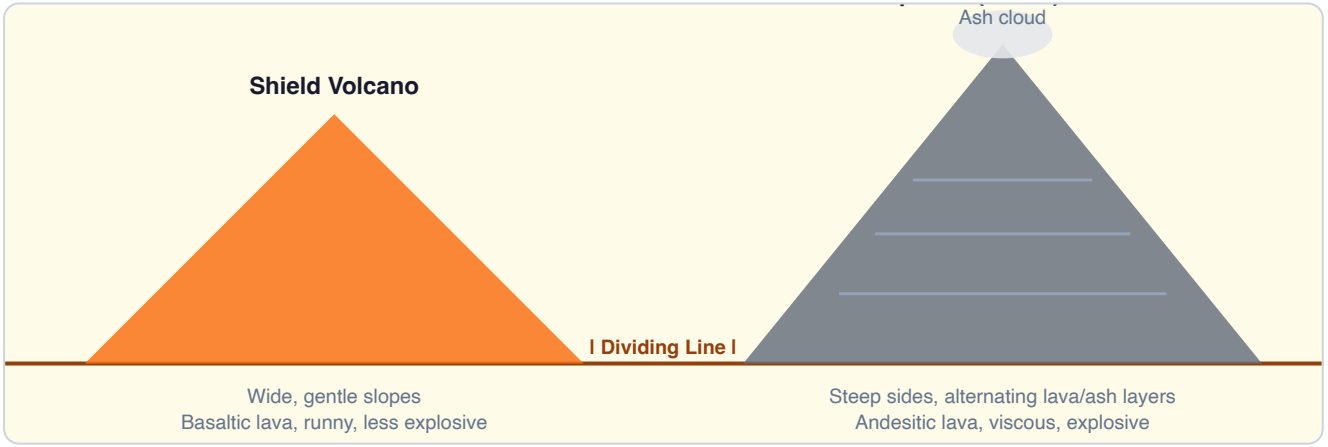
# 5. Conservative (Transform) Plate Boundaries

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- Plates slide past each other horizontally — no crust created or destroyed.
- **No volcanic activity** (no subduction, no magma generation).
- Friction builds between plates — when released, produces **very powerful earthquakes** (shallow focus, extremely destructive near populated areas).
- **San Andreas Fault, California** — North American plate vs Pacific plate (Pacific moving NW at ~5cm/year). San Francisco earthquake 1906 (estimated 7.9 magnitude). Los Angeles earthquake 1994 — Northridge, 6.7 magnitude, 57 deaths, $20bn damage.
- The Dead Sea Transform fault (Jordan/Israel/Palestine) is another example.

## 6. Volcanoes




| Feature | Shield Volcano | Composite / Stratovolcano |
|---|---|---|
| Shape | Wide, gently sloping (1–10°) | Steep, cone-shaped (30–40°) |
| Lava type | Basaltic — low viscosity, runny | Andesitic/rhyolitic — high viscosity, thick |

| | | |
|---|---|---|
| Eruption style | Frequent, less explosive (effusive) | Infrequent but explosive (pyroclastic) |
| Hazards | Lava flows (slow-moving) | Pyroclastic flows, ash fall, lahars, bombs |
| Boundary type | Constructive / Hot spot | Destructive (subduction zones) |
| Examples | Mauna Loa (Hawaii), Kilauea, Piton de la Fournaise (Réunion) | Mt St Helens (USA), Pinatubo (Philippines), Merapi (Indonesia), Etna (Italy) |

## Volcanic Hazards

- **Lava flows** — slow-moving (most people can escape), but destroy property and farmland.
- **Pyroclastic flows** — fast-moving (up to 700km/h), superheated (400–700°C) mixtures of gas, ash, and rock. Most deadly volcanic hazard. Killed ~29,000 in Mt Pelée, Martinique (1902).
- **Ash fall** — suffocates crops, collapses buildings, disrupts air travel (Eyjafjallajokull 2010 — 100,000 flights cancelled).
- **Lahars** — volcanic mudflows. Melted snow/ice mixes with ash. Armero, Colombia (1985): lahar from Nevado del Ruiz killed 23,000.
- **Volcanic gases** — $SO_2$, $CO_2$ toxic; can cause acid rain.
- **Tsunamis** — underwater eruptions can trigger tsunamis.

## Benefits of Volcanoes

- **Fertile soils** — volcanic ash weathered to nutrient-rich soils. Java (Indonesia) is one of the world's most densely populated areas due to highly fertile volcanic soils.
- **Geothermal energy** — Iceland generates 25% of electricity from geothermal energy — heats 90% of homes.
- **Tourism** — Mt Etna, Kilauea, Yellowstone attract millions of visitors.
- **Minerals** — valuable minerals (copper, gold, sulphur) deposited around volcanoes.

# 7. Earthquakes

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**Focus** — the point inside the Earth where an earthquake originates (where rock fractures/slips). Also called the hypocenter.
**Epicentre** — the point on the Earth's surface directly above the focus. Most severe shaking occurs here.
**Seismic waves** — energy waves that travel outward from the focus.

## Types of Seismic Waves

- **P-waves (Primary)** — compression waves; travel through solid and liquid; fastest; arrive first on seismograph.
- **S-waves (Secondary)** — shear waves; only travel through solid; slower; cannot travel through Earth's liquid outer core.
- **Surface waves (Love and Rayleigh)** — travel along Earth's surface; slowest but most destructive — cause most of the ground shaking and building damage.

## Measuring Earthquakes

- **Richter Scale** — measures magnitude (energy released). Logarithmic: each whole number = 10x greater amplitude (about 32x more energy). 0–2 (not felt) → 5–6 (moderate damage) → 7+ (severe) → 9+ (extreme, rare).
- **Moment Magnitude Scale (Mw)** — now preferred by scientists; similar logarithmic scale but more accurate for large earthquakes.
- **Mercalli Scale** — measures intensity (damage and shaking felt at a location). Rated I–XII. Subjective but useful where seismometers are absent.

## Effects of Earthquakes

| Primary Effects (immediate, caused directly by shaking) | Secondary Effects (caused by primary effects) |
|---|---|
| Buildings and bridges collapse | Tsunamis (if submarine earthquake) |
| Ground shaking and cracking | Landslides and avalanches |

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| | |
|---|---|
| People killed/injured by collapsing structures | Liquefaction — waterlogged soil behaves like liquid, buildings sink (Christchurch NZ 2011) |
| Roads and railways buckle/break | Disease outbreaks (cholera, typhoid) from damaged sanitation |
| Fires from broken gas pipes | Homelessness → disease, exposure, mental health impacts |

## 8. Tsunamis

A tsunami is a series of powerful ocean waves caused by a large underwater disturbance (earthquake, landslide, volcanic eruption). Most tsunamis are caused by **submarine earthquakes** — where the seafloor suddenly shifts vertically, displacing a vast column of water.

- **Deep water:** wave speed ~800 km/h, wavelength 100–200 km, amplitude just 1–2m (often unnoticed at sea).
- **Shallow water (approaching coast):** speed slows dramatically but wave height increases enormously ("shoaling") — can reach 30–40m.
- Warning signs: sea withdraws rapidly from shore (draw-back) before the wave arrives.

### 2004 Indian Ocean Tsunami (Boxing Day Tsunami)

- **Cause:** 9.1 magnitude submarine earthquake (destructive boundary — Indo-Australian plate subducting under Burma plate) 250km off Sumatra, Indonesia.
- **Scale:** 230,000+ deaths across 14 countries (Indonesia, Sri Lanka, India, Thailand — worst). 1.7 million displaced.
- **Wave height:** Up to 30m in Sumatra; 6–9m in Sri Lanka/India.
- **Speed:** ~800 km/h across Indian Ocean — reached Sri Lanka in ~2 hours, Africa in ~7 hours.
- **Response:** Led to creation of Indian Ocean Tsunami Warning System (2006). Pacific Tsunami Warning Centre (PTWC) already existed since 1949.

## 9. Case Study 1: Nepal Earthquake, April 2015 (LIC)

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### Nepal Earthquake — 25 April 2015 | 7.8 Magnitude

**Location:** 80km NW of Kathmandu, shallow focus (15km depth), continent-continent destructive boundary (Indo-Australian plate colliding with Eurasian plate — Himalayas).

## Primary Effects

- 9,000+ deaths; 23,000+ injured.
- 600,000+ homes destroyed (mostly mud-brick construction).
- 8,000+ schools and 1,000 health facilities damaged.
- Historic Durbar Square, Kathmandu (UNESCO World Heritage Site) — temples collapsed.
- Dharahara Tower (19th century) collapsed, killing 180 people sheltering nearby.

## Secondary Effects

- Massive landslides — Langtang village buried (350 people), trekking routes blocked.
- Avalanches on Everest — 19 climbers killed at Base Camp; worst Everest disaster at the time.
- 3 million people left homeless — displacement raised risk of disease (cholera, diarrhoea) as sanitation systems destroyed.
- Crops destroyed ahead of harvest — food insecurity.
- Tourism, a major source of income and jobs, collapsed in the short term.

## Why Was Impact So Severe? (LIC factors)

- **Poor building quality** — most rural housing built from unmortared stone or mud brick. No earthquake-resistant design. Urban buildings often poorly constructed despite regulations.
- **Remote mountainous terrain** — roads blocked by landslides; helicopter access only; aid reached many areas weeks late.
- **Population density** — Kathmandu Valley densely populated (1.5 million+), high exposure.
- **Poverty** — Nepal is one of Asia's poorest nations; GNI per capita was under $1,000 at the time. Limited government capacity to prepare or respond.

## Responses

- **Immediate:** International aid poured in — India first responder (NDRF teams). USA, UK, China, Pakistan, EU — $4.4bn pledged internationally. UN OCHA coordinated. Nepali Army deployed

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search and rescue.

- **Short-term issues:** Aid bottleneck at Kathmandu airport (only one runway); looting of relief supplies; corruption allegations; political disputes delayed relief distribution.
- **Long-term:** "Build Back Better" reconstruction programme — earthquake-resistant design. By the fifth anniversary (2020), many households were still waiting to complete rebuilding. Slow bureaucracy, land disputes, funding mismanagement criticised heavily by survivors and NGOs.

# 10. Case Study 2: Japan Earthquake and Tsunami, March 2011 (HIC)

### Tohoku Earthquake and Tsunami — 11 March 2011 | 9.0 Magnitude

**Location:** 70km off Sendai coast, NE Japan. Destructive boundary — Pacific plate subducting under North American/Eurasian plate (Japan Trench). Strongest earthquake ever recorded in Japan; 4th strongest globally since 1900.

## Primary Effects

- 15,900+ deaths (mostly by tsunami, not earthquake — buildings withstood shaking well).
- 130,000+ buildings destroyed; 1 million+ damaged.
- Fukushima Daiichi nuclear power plant — tsunami overwhelmed sea defences (a ~14 m wave vs defences designed for a tsunami of about 6 m), causing cooling systems to fail — nuclear meltdown in 3 reactors. Level 7 nuclear emergency (same as Chernobyl).
- Tsunami: heights up to 40m at some coastal points; travelled up to 10km inland; swept away entire coastal towns (Ishinomaki, Rikuzentakata).

## Secondary Effects

- **Nuclear disaster:** 154,000 people evacuated from 20km exclusion zone around Fukushima. Long-term radiation contamination of soil and sea. As of 2023, exclusion zones still partially in place.
- Global nuclear energy debate reignited — Germany accelerated nuclear phase-out; Japan shut all 50 reactors for safety checks.

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- Supply chain disruption — Japan produces 60% of world's silicon wafers for semiconductors; global electronics industry affected for months.
- Economic cost: ~$235bn — most expensive natural disaster in history.

### Why Japan Coped Better (HIC factors)

- **Earthquake-resistant buildings** — strict building codes enforced since 1981 (and revised after 1995 Kobe quake): cross-bracing, shock absorbers, base isolators, automatic gas shut-off.
- **Early warning system** — Japan Meteorological Agency issued an earthquake early warning seconds after the quake was detected — giving up to about a minute before strong shaking reached Tokyo; automated systems shut down trains and industrial machinery.
- **Tsunami warning** — coastal warning issued within 3 minutes; evacuation sirens activated.
- **Regular drills** — Disaster Prevention Day (1 September) — nationwide earthquake drills every year since 1960.
- **Organised emergency response** — Self-Defence Forces (100,000 deployed), pre-positioned emergency supplies, efficient hospital system.

### BUT — Limitations Even for an HIC

- Tsunami overwhelmed sea walls designed for smaller events — underestimation of risk.
- Nuclear disaster was unexpected and unprecedented — containment plans inadequate.
- Evacuation traffic jams reduced escape rates in some coastal areas.
- Elderly population (Japan has world's oldest population) — struggled to evacuate quickly.

## Comparison: Nepal (LIC) vs Japan (HIC)

| Factor | Nepal 2015 (LIC) | Japan 2011 (HIC) |
|---|---|---|
| Magnitude | 7.8 | 9.0 (much larger) |
| Deaths | 9,000+ | 15,900+ (mainly from tsunami) |
| Building quality | Mud brick, poorly built | Earthquake-resistant to modern codes |

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| | | |
|---|---|---|
| Early warning | No effective system | Earthquake early warning (up to ~1 minute in Tokyo); tsunami alert in 3 min |
| Emergency response | Slow, disorganised, aid bottlenecks | Fast, organised, 100,000 military deployed |
| Long-term recovery | Still incomplete 5+ years later | Most areas rebuilt within 3–5 years |
| Main factor for deaths | Building collapse + LIC poverty | Tsunami inundation (not earthquake) |

## 11. Managing Tectonic Hazards

AQA groups management into **monitoring, prediction, protection and planning**, all of which aim to reduce the risk from tectonic hazards. International aid supports the response after an event.

### Monitoring and Prediction

- **Volcanoes** — ground swelling (tiltmeters), increased seismic activity (seismometers), gas emissions ($SO_2$ and $CO_2$ levels rise before eruptions), thermal cameras detecting heat changes. Mt Pinatubo (1991) — successful 2-week warning: 60,000 people evacuated, saved thousands of lives.
- **Earthquakes** — cannot be predicted with sufficient accuracy for evacuation. Seismometers detect P-waves seconds before damaging S/surface waves arrive (seconds to ~1 minute warning). Long-term probabilistic hazard mapping identifies high-risk areas.

### Protection (Engineering)

- **Earthquake-resistant buildings:** cross-bracing (steel frames), base isolators (rubber pads between building and foundations absorb shaking), counterweights (dampers on tall buildings — Taipei 101 has a 660-tonne pendulum), automatic gas shut-off, fire-resistant materials.
- **Tsunami sea walls:** Japan's Kamaishi sea wall (about 2km long, 63m deep foundations, 8m above sea level) — largest in the world at construction. The 2011 tsunami overtopped it but it reduced wave height and gave more evacuation time.

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- **Breakwaters and coastal barriers** — Japan invested $12bn in tsunami sea walls along 400km of coastline after 2011.

### Planning

- Land-use zoning — restricting development in high-risk zones near faults or low-lying coasts.
- Evacuation routes — clearly signed, regularly rehearsed.
- Regular drills — Japan (annual), California (Great ShakeOut — 10 million+ participants).
- Community education — knowing what to do in the first 72 hours (the critical window for rescuing survivors).

### Aid

International response critical for LICs (Nepal, Haiti 2010). UNOCHA coordinates. Issues: logistics, corruption, dependency. Long-term reconstruction aid must focus on building back safer, not just rebuilding the same vulnerable structures.

**Exam Tip:** For "evaluate management strategies" questions, use the framework: Prediction is *difficult for earthquakes* (but useful for volcanoes); Protection is *effective but expensive* (only HICs can fully implement it); Planning is *most cost-effective* and works in both HICs and LICs; Aid is *vital in the short term* but must avoid dependency. Always refer to a named example for each strategy.

## 12. Why Do People Continue to Live in Tectonic Hazard Zones?

| Reason | Explanation and example |
|---|---|
| Fertile soils | Volcanic ash weathers into mineral-rich soils that give high crop yields — e.g. slopes of Mt Etna (Sicily) produce citrus fruit, olives and wine grapes; Java (Indonesia) supports very dense farming populations. |
| Geothermal energy | Iceland uses geothermal heat for around 25% of its electricity and about 90% of home heating — cheap, renewable energy. |
| Tourism | Volcanoes and geothermal features attract visitors, creating jobs — e.g. Iceland's Blue Lagoon, Mt Etna, Yellowstone. |

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| | |
|---|---|
| Minerals | Valuable minerals (sulphur, copper, gold, diamonds) are found near volcanoes and are mined for income. |
| Family and work ties | People have lived there for generations, own land and homes, and have jobs and family nearby — moving is expensive and disruptive. |
| Confidence in management | In HICs such as Japan or the USA, people trust building codes, monitoring and warning systems to keep them safe. |
| Perception of risk | Eruptions and large earthquakes are infrequent, so people underestimate the risk or believe "it won't happen in my lifetime". |

**Exam Tip:** "Suggest why people continue to live in areas at risk from tectonic hazards" is a classic 4-mark question. Give two reasons, each developed with a consequence ("...which means...") and, ideally, a named place.

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